

Code No: 115EH**JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD****B. Tech III Year I Semester Examinations, November - 2015****OPERATING SYSTEMS****(Common to CSE, IT)****Time: 3 hours****Max. Marks: 75****Note:** This question paper contains two parts A and B.

Part A is compulsory which carries 25 marks. Answer all questions in Part A. Part B consists of 5 Units. Answer any one full question from each unit. Each question carries 10 marks and may have a, b, c as sub questions.

PART - A (25 Marks)

- 1.a) What do you mean by Virtual Memory? [2]
- b) Define multitasking. [3]
- c) What are the two separate and potentially independent characteristics embodied in the concept of process? [2]
- d) Describe about race condition. [3]
- e) What is the difference between page and segment? [2]
- f) What do you mean by Thrashing? [3]
- g) Define three blocking methods. [2]
- h) What delay elements are involved in disk read or write? [3]
- i) How can the hold-and-wait condition be prevented? [2]
- j) What are the necessary and sufficient conditions to occur deadlock? [3]

PART - B (50 Marks)

- 2.a) What system calls have to be executed by a command interpreter or shell in order to start a new process? Explain briefly.
- b) What are the main differences between operating systems for mainframe computers and personal computers? [5+5]

OR

- 3.a) What is the relationship between a guest operating system and a host operating system in a system like VMware? What factors need to be considered?
- b) Define Essential properties of Distributed Operating Systems. [5+5]

- 4.a) Explain process states.
- b) Explain the concept of semaphores. Illustrate with an example. [5+5]

OR

- 5.a) Give Peterson solution for critical section problem.
- b) Consider the following set of process, with the length of the CPU burst given in milliseconds. [3+7]

Process	Burst Time	Priority
P1	10	3
P2	1	1
P3	2	3
P4	1	4
P5	5	2

The processes are assumed to have arrived in the order P₁, P₂, P₃, P₄, P₅, all at time 0. What is the turnaround time of each process by applying priority scheduling algorithm?

- 6.a) Differentiate Internal fragmentation and external fragmentation.
b) What is Belady's anomaly? Explain with one example. [5+5]

OR

7. Consider the following page reference strings:
1,2,3,4,2,1,5,6,2,1,2,3,7,6,3,2,1,2,3,6.
How many page faults would occur for the following replacement algorithm, assuming three, four frames? Remember that all frames are initially empty, so your first unique pages will cost one fault each.
a) LRU replacement
b) Optimal replacement. [5+5]

- 8.a) Write about Swap space management.
b) Suppose that a disk drive has 5,000 cylinders, numbered 0 to 4999. The drive is currently serving a request at cylinder 143, and the previous request was at cylinder 125. The queue of pending requests, in FIFO order, is: **86, 1470, 913, 1774, 948, 1509, 1022, 1750, 130**. Starting from the current head position, what is the total distance (in cylinders) that the disk arm moves to satisfy all the pending requests for each of the SSTF disk scheduling algorithms? [4+6]

OR

9. Explain in detail about different file allocation methods. [10]
10. List the conditions that must be present for deadlock to occur and for each condition give brief example or reason that illustrate a disadvantage in preventing the condition. [10]

OR

11. State whether you agree or disagree with the following claim:
The Banker's algorithm may fail to avoid deadlock if a resource breaks down. Justify your answer. [10]

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Code No: 115EH**JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD****B. Tech III Year I Semester Examinations, November/December - 2016****OPERATING SYSTEMS****(Common to CSE, IT)****Time: 3 hours****Max. Marks: 75****Note:** This question paper contains two parts A and B.

Part A is compulsory which carries 25 marks. Answer all questions in Part A. Part B consists of 5 Units. Answer any one full question from each unit. Each question carries 10 marks and may have a, b, c as sub questions.

PART - A**(25 Marks)**

- 1.a) Distinguish between symmetric and asymmetric multi processor systems. [2]
- b) Define the essential properties of Interactive operating systems and Network operating systems. [3]
- c) What is a dispatcher process? Explain its role. [2]
- d) Describe the differences between preemptive scheduler and non-preemptive scheduler. [3]
- e) What is the need of dynamic loading and dynamic linking? [2]
- f) Explain the differences between internal fragmentation and external fragmentation. [3]
- g) Define mounting. What is the need for mounting in a file system? [2]
- h) What are the typical operations that can be performed on directory? [3]
- i) What is deadlock? What is starvation? How do they differ from each other? [2]
- j) What are the various methods for protection and access control? [3]

PART - B**(50 Marks)**

- 2.a) Explain briefly about virtual machines and micro Kernels.
- b) Define operating system goals from user's view and system's view. [5+5]

OR

3. What are the major activities of an operating system with regard to file management? Explain them briefly with their supporting system calls. [10]

4. Following is the snapshot of a CPU

Process	CPU Burst	Arrival Time
P1	75	0
P2	40	10
P3	25	10
P4	20	80
P5	45	85

Draw the Gantt chart and calculate the turnaround time and waiting time of the jobs for FCFS (First Come First Served), SJF (Shortest Job First), SRTF (Shortest Remaining Time First) and RR (Round Robin with time quantum 15) scheduling algorithms. [10]

OR

5. What is a monitor? Explain how dining philosopher's problem is solved using monitors with example pseudo code. [10]

6. Explain the common techniques for structuring the page table. [10]

OR

7.a) Consider a swapping system in which memory consists of the following hole sizes in memory order: 10 KB, 4 KB, 20 KB, 18 KB, 7 KB, 9 KB, 12 KB, and 15 KB. Which hole is taken for successive segment requests of: (i) 12 KB (ii) 10 KB (iii) 9 KB for first fit, best fit, worst fit, and next fit approaches.

b) Explain briefly about LFU Page replacement algorithm. [5+5]

8. Explain the following with relevant diagrams:

a) Two level directory structure.

b) Acyclic-graph directory structure. [5+5]

OR

9.a) Explain any two methods used to protect user files with examples.

b) Discuss the log structured file system implementation. [5+5]

10. Consider the following snapshot of a system:

Processes	Allocation	Max	Available
	A B C D	A B C D	A B C D
P0	0 0 1 2	0 0 1 2	2 1 0 0
P1	2 0 0 0	2 7 5 0	
P2	0 0 3 4	6 6 5 6	
P3	2 3 4 5	4 3 5 6	
P4	0 3 3 2	0 6 5 2	

Answer the following questions using the banker's algorithm:

a) What is the content of the matrix Need?

b) Is the system in a safe state? Why?

c) Is the system currently deadlocked? Why or why not?

d) Which process, if any, or may become deadlocked if this whole request is granted immediately? [2+3+2+3]

OR

11. In the capability-based system, describe the techniques, which can be used to protect the capabilities from unauthorized modification. [10]

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Code No: 115EH**JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD****B. Tech III Year I Semester Examinations, March - 2017****OPERATING SYSTEMS****(Common to CSE, IT)****Time: 3 hours****Max. Marks: 75****Note:** This question paper contains two parts A and B.

Part A is compulsory which carries 25 marks. Answer all questions in Part A. Part B consists of 5 Units. Answer any one full question from each unit. Each question carries 10 marks and may have a, b, c as sub questions.

PART - A**(25 Marks)**

- 1.a) Define system call. [2]
- b) Differentiate between multi programming and multi processing. [3]
- c) What is critical section problem? [2]
- d) How are processes managed in LINUX? [3]
- e) Differentiate between logical virtual and physical address. [2]
- f) Explain directory structure. [3]
- g) Give different RAID levels. [2]
- h) Discuss about swap space management. [3]
- i) Differentiate between unsafe state and dead lock state. [2]
- j) How access rights are revoked? [3]

PART - B**(50 Marks)**

- 2.a) Briefly explain typical functions of an Operating-System Kernel.
 - b) What resources are used when a thread is created? How do they differ from those used when a process is created? [5+5]
- OR**
- 3.a) What are the different types of operating systems? Explain them in detail.
 - b) What are the main characteristics of Real Time Operating System? [5+5]
4. Discuss readers/writers problem and give solution by using semaphores where readers have priority. [10]
- OR**
5. Construct the Gantt chart for a) Shortest job first b) Round Robin with q=3 c) Round robin with q=4 d) shortest remaining time first scheduling algorithms for the following. [10]

Process	P1	P2	P3	P4	P5
Arrival time	0	0	2	1	3
CPU Burst Time (in ms)	10	6	12	8	5

6. Explain how protection can be ensured using paging? [10]

OR

7.a) A process refers to 5 pages, A, B, C, D, and E in the order- A; B; C; D; A; B; E; A; B; C; D; E. If the page replacement algorithm is LRU, calculate the number of page faults with empty frames of size 4?

b) Explain the terms in Memory Partitioning with examples:
i) Fixed Partitioning ii) Dynamic partitioning. [5+5]

8. Suppose the head of a moving head disk with 200 tracks, numbered 0 to 199, is currently serving a request at track 143 and has just finished a request at track 125. If the queue of requests is kept in FIFO order: 86, 147, 91, 177, 94, 150, 102, 175, 130. What is the total head movement to satisfy these requests for the following disk scheduling algorithms?

(a) FCFS (b) SCAN (c) SSTF (d) C- SCAN [10]

OR

9.a) What is a Directory? Write short note on Directory implementation.

b) Explain about linked allocation method of a file. [5+5]

10. A system has 3 devices D1, D2 and D3 and 3 processes P1, P2, and P3. P1 is holding D1 and waiting for D3. P2 is holding D2 and waiting for D1. P3 is holding D3 and waiting for D2. Draw resource allocation graph and wait-for graph. Is the system in deadlock state or not? Explain. [10]

OR

11.a) Explain about capability based systems.

b) Discuss about revocation of access rights. [5+5]

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Code No: 115EH**JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD****B.Tech III Year I Semester Examinations, November/December - 2018****OPERATING SYSTEMS****(Common to CSE, IT)****Time: 3 hours****Max. Marks: 75****Note:** This question paper contains two parts A and B.

Part A is compulsory which carries 25 marks. Answer all questions in Part A. Part B consists of 5 Units. Answer any one full question from each unit. Each question carries 10 marks and may have a, b, c as sub questions.

PART - A**(25 Marks)**

- 1.a) What are the goals of operating system? [2]
- b) What is an Operating system? What are the various OS Components? [3]
- c) What is critical section problem? [2]
- d) What is preemptive Scheduling? How is it different from non-preemptive scheduling? [3]
- e) What is Virtual Memory? Why is it required? [2]
- f) Compare LRU and Optimal Replacement Algorithms. [3]
- g) What are the files attributes up dated during file creation? [2]
- h) What are the methods for accessing the file? [3]
- i) Define deadlock. [2]
- j) Write down the principles of protection. [3]

PART - B**(50 Marks)**

2. Describe evolution of operating system in detail. [10]
- OR**
3. What is the need for system calls? Explain the types of system calls provided by an operating system with respect to memory management. [10]
4. Explain FCFS, RR and SJF scheduling algorithm with illustrations. [10]
- OR**
5. Explain about multiple-processor scheduling and real time scheduling. [10]
- 6.a) What is the need of demand paging? Explain briefly.
- b) Explain in detail about segmentation on with paging technique. [5+5]
- OR**
- 7.a) Explain in detail about paging technique.
- b) Explain in detail Contiguous Memory Allocation. [5+5]
8. Explain in detail about file system structure and implementation. [10]
- OR**
9. Explain in detail about free space management. [10]

10. How can deadlock be detected and recovered? Explain in detail with relevant example. [10]

OR

11.a) What is access matrix? What are various methods to implement it?

b) Explain Capability-Based Protection system. [5+5]

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Code No: 115EH**JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD****B. Tech III Year I Semester Examinations, May/June - 2019****OPERATING SYSTEMS****(Common to CSE, IT)****Time: 3 hours****Max. Marks: 75****Note:** This question paper contains two parts A and B.

Part A is compulsory which carries 25 marks. Answer all questions in Part A. Part B consists of 5 Units. Answer any one full question from each unit. Each question carries 10 marks and may have a, b, c as sub questions.

PART - A**(25 Marks)**

- 1.a) Distinguish between the client-server and peer-to-peer models of distributed systems. [2]
- b) Define the essential properties of Time-sharing systems and Clustered systems. [3]
- c) What is a scheduler process? Explain its role. [2]
- d) Describe the differences between short-term scheduler and long-term scheduler. [3]
- e) What is the need of mapping between logical address and physical address? [2]
- f) What is thrashing? [3]
- g) What is file system mounting? [2]
- h) Discuss the objectives for a file management system. [3]
- i) What are the four conditions required for deadlock to occur? [2]
- j) What are the principles of protection? [3]

PART - B**(50 Marks)**

- 2.a) Explain various system calls for process management.
 - b) With a neat diagram describe how a modern computer works? [5+5]
- OR**
- 3.a) Explain about the structure of OS.
 - b) Explain about the services of operating system. [5+5]

4. Following is the snapshot of a CPU

Process	CPU Burst	Arrival Time
P1	10	0
P2	29	1
P3	03	2
P4	07	3

Draw the Gantt chart and calculate the turnaround time and waiting time of the jobs for FCFS (First Come First Served), SJF (Shortest Job First), SRTF (Shortest Remaining Time First) and RR (Round Robin with time quantum 10) scheduling algorithms. [10]

OR

5. What is a semaphore? Explain how producer-consumer problem is solved using semaphores with example pseudo code. [10]

6. Give the detailed description of hardware implementation of a page table with translation Look-Aside Buffer. [10]

OR

- 7.a) Consider a swapping system in which memory consists of the following hole sizes in memory order: 12 KB, 4 KB, 24 KB, 15 KB, 9 KB, 7 KB, 10 KB, and 11 KB. Which hole is taken for successive segment requests of : (i) 14 KB (ii) 8 KB (iii) 5 KB for first fit, best fit, worst fit, and next fit approaches.
- b) Explain briefly about optimal Page replacement algorithm. [5+5]

8. Explain the following with relevant diagrams:

a) Single level directory structure.

b) Tree-structured directory structure. [5+5]

OR

- 9.a) Explain about the free space management.

- b) Explain about the linear list and hash table data structures to implement a directory. [5+5]

10. Consider the following snapshot of a system:

Processes	Allocation	Max	Available
	A B C D	A B C D	A B C D
P0	0 0 1 2	0 0 1 2	1 5 2 0
P1	1 0 0 0	1 7 5 0	
P2	1 3 5 4	2 3 5 6	
P3	0 6 3 2	0 6 5 2	
P4	0 0 1 4	0 6 5 6	

Answer the following questions using the banker's algorithm:

a) What is the content of the matrix Need?

b) Is the system in a safe state?

c) If a request from process P1 arrives for (0, 4, 2, 0) can the request be granted immediately? [3+3+4]

OR

11. What is meant by authentication? Why simple password protection is the most common authentication scheme in use today? Discuss the weakness inherent in the password protection scheme. [10]

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